Squared Digit Sums Are Equidistributed Modulo Their Base

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Abstract

For every integer base $b \geq 2$, each residue of the squared digit sum modulo b has natural density $1/b$. The increasing sequence of positions of any one residue therefore has asymptotic slope b. This proves the all-base density and slope conjecture stated in OEIS A079495, including composite bases. We also give exact finite digit recurrences for every counting function and every sequence term, and an exact count on ternary length blocks.

1 The question and the full statement

For the base-b digits of $n \geq 0$, let $s_b(n)$ be the sum of their squares, with $s_b(0) = 0$. Write

$$C_{b,r}(N) = \#\{0 \leq n < N : s_b(n) \equiv r \pmod b\}, \qquad 0 \leq r < b.$$

OEIS A079495 [1] lists the positions of residue zero for $b = 3$, starting with zero. Its comments conjecture density $1/b$ and positional slope b in every base. The related digit-sum entry A079498 [2] already contains a proof for ordinary digit sums; the present argument treats squared digits.

Theorem 1. *For every fixed $b \geq 2$ and $0 \leq r < b$,*

$$C_{b,r}(N) = \frac{N}{b} + o(N) \qquad (N \longrightarrow \infty).$$

If $a_{b,r}(j)$ is the jth nonnegative integer with residue r, indexed by $j \geq 1$, then

$$\lim_{j\to\infty} \frac{a_{b,r}(j)}{j} = b.$$

Proof. Put $\zeta = \exp(2\pi i/b)$ and

$$\lambda_\ell = \frac{1}{b}\sum_{d=0}^{b-1} \zeta^{\ell d^2}, \qquad 1 \leq \ell < b, \qquad \rho = \max_{1\leq \ell<b} |\lambda_\ell|.$$

Every summand has modulus one. The terms for $d = 0$ and $d = 1$ are 1 and ζ^ℓ, which are distinct. Strictness in the triangle inequality therefore gives $\rho < 1$. This uses no primality assumption on b.

For a complete block of k unrestricted digits, character orthogonality gives the exact count

$$C_{b,r}(b^k) = \frac{b^k}{b}\left(1 + \sum_{\ell=1}^{b-1} \zeta^{-\ell r}\lambda_\ell^k\right). \tag{1}$$

Its discrepancy from b^k/b is at most $(b-1)b^{k-1}\rho^k$. More generally a block $[Ab^k, (A+1)b^k)$ fixes the leading digits of A and has k unrestricted trailing digits. Adding $s_b(A)$ merely shifts the residue, so the same discrepancy bound holds for every such block. For $k = 0$ use the bound $(b-1)/b$ directly.

The base-b expansion of N decomposes $[0, N)$ into at most $b-1$ such blocks of each size b^k, $0 \le k \le L$, where $L = \lfloor \log_b N \rfloor$. Consequently

$$\left| C_{b,r}(N) - \frac{N}{b} \right| \le \frac{(b-1)^2}{b} \sum_{k=0}^{L} (b\rho)^k, \tag{2}$$

where the $k = 0$ summand is one, also when $\rho = 0$. If $b\rho < 1$ this is bounded; if $b\rho = 1$ it is $O_b(\log(N+1))$; if $b\rho > 1$ it is $O_b(N^{1+\log_b \rho})$. In the last case the exponent is strictly smaller than one. All three cases prove density $1/b$.

In particular each residue occurs infinitely often. At its jth position, $C_{b,r}(a_{b,r}(j)+1) = j$, so $j = (a_{b,r}(j)+1)/b + o(a_{b,r}(j)+1)$. Division and inversion give the asserted slope. □

2 Exact finite counts and sequence terms

Let $V_k(r) = C_{b,r}(b^k)$. Entirely in integer arithmetic,

$$V_0(r) = \begin{cases} 1, & r = 0, \\ 0, & r \ne 0, \end{cases} \qquad V_{k+1}(r) = \sum_{d=0}^{b-1} V_k(r - d^2 \bmod b). \tag{3}$$

To count below an arbitrary N, read its digits from left to right. At a position having digit c and k positions to its right, let u be the squared sum of its already fixed prefix. For each $0 \le d < c$, add

$$V_k(r - u - d^2 \bmod b).$$

Then fix the actual digit c and update $u \leftarrow u + c^2$. The disjoint first-differing-digit cases partition $[0, N)$, proving this exact digit formula. Leading zeros cause no change in s_b.

These counts also compute every sequence term: take a length k with $V_k(r) \ge j$ and perform binary search for the least N with $C_{b,r}(N) \ge j$; then $a_{b,r}(j) = N - 1$. Theorem 1 proves that such a finite k always exists. This is a finite digit algorithm, not an elementary closed expression for the jth position.

Corollary 1. *For base three, the number of A079495 terms below 3^k is*

$$V_k(0) = \begin{cases} 3^{k-1}, & k \text{ odd}, \\ \left(3^k + 2(-3)^{k/2}\right)/3, & k \text{ even}. \end{cases}$$

The even case includes $k = 0$, when the value is one.

Proof. For $\zeta = \exp(2\pi i/3)$, the digit squares are $0, 1, 1$ modulo three, so $\lambda_1 = (1 + 2\zeta)/3 = i/\sqrt{3}$ and $\lambda_2 = -i/\sqrt{3}$. Substitute these values into (1) with $r = 0$. □

3 Scope and independent checks

The accompanying `verify.py` compares every integer-prefix count for bases 2 through 16 and $0 \le N \le 5000$ with direct digit extraction. It compares all 63 displayed A079495 terms, checks the exact ternary block counts through length 100 by integer convolution, and independently checks sequence selection for each residue in each tested base. The all-base and asymptotic claims follow from the proof, not finite numerical ratios. No literature priority or OEIS acceptance is asserted.

References

[1] C. Alves, OEIS A079495, https://oeis.org/A079495, accessed October 26, 2026.

[2] C. Alves, OEIS A079498, including M. Dekking's April 18, 2019 proof for ordinary digit sums, https://oeis.org/A079498, accessed October 26, 2026.